

CorBin-FL: A Differentially Private Federated Learning Mechanism using Correlated Quantization

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ABSTRACT Federated learning (FL) has emerged as a promising framework for distributed machine learning. However, FL faces challenges in balancing privacy guarantees, communication efficiency, and overall model accuracy. This work introduces CorBin-FL, a privacy mechanism that employs correlated binary stochastic quantization to achieve differential privacy. The approach leverages secure multi-party computation techniques to enable clients to perform correlated quantization of their local model updates without compromising individual privacy. In the proposed implementation, paired clients establish a shared secret key through the server facilitated Diffie-Hellman protocol and exchange a compact PRG seed to generate shared randomness locally, resulting in negligible additional communication overhead comparing to independent quantization. Theoretical analysis is provided showing that CorBin-FL achieves parameter-level local differential privacy (PLDP), and asymptotically optimizes the privacy-utility trade-off between the mean square error utility measure and the PLDP privacy measure. Furthermore the AugCorBin-FL is proposed, which is an extension of CorBin-FL that, in addition to PLDP, achieves central differential privacy (CDP) guarantees. For both mechanisms, bounds on privacy parameters and mean squared error performance measures are derived. Empirical evaluations across computer vision (MNIST, CIFAR-10, FEMNIST), sequence prediction (Shakespeare), and sentiment analysis (Sentiment140) datasets demonstrate consistent accuracy improvements over existing privacy mechanisms under equal privacy budgets.

INDEX TERMS Correlated quantization, federated learning, differential privacy.

I. INTRODUCTION

DISTRIBUTED stochastic gradient descent in general, and federated learning (FL) in particular, are foundational concepts in modern machine learning with wide-ranging applications [1], [2]. Federated learning inherently provides a degree of privacy by not requiring clients to share raw data. This is further enhanced when combined with secure aggregation techniques [3]. However, recent works have shown that both the aggregated model updates at each round and the final model weights may leak information about the training data, and even potentially enable the reconstruction of sensitive data [4].

To provide quantifiable guarantees against data leakage, the concept of differential privacy (DP) has emerged as a crucial framework. At a high level, DP requires that the output of an (aggregation) mechanism remains essentially

unchanged when any single individual's data is added to or removed from the input dataset. Several different notions of DP have been considered in FL. User-level central DP (UCDP) requires that the aggregate model at each round does not reveal the participation of any individual client in that round of training [5]–[7]. Sample-level central DP (SCDP) is more stringent and requires that the aggregate model does not reveal information about usage of any single training sample [8], [9]. Local DP (LDP) requires that an eavesdropper gaining access to the local update being transmitted by one of the clients in uplink communication does not gain information about any single training samples used by the client [10]–[12]. Prior works have considered injecting noise into the updates to achieve DP. These include the Gaussian mechanism, Laplacian mechanism, and the Discrete Gaussian mechanism, among others [6], [13], [14].

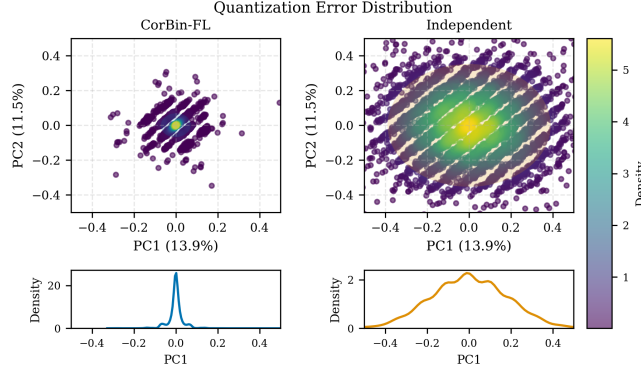


FIGURE 1. PCA visualization of quantization errors for CorBin-FL (left) and LDPFL (right) on MNIST with 50 clients and $\epsilon_p = 0.5$. Each point represents the PCA-projected quantization error from 10,000 sampled model parameters over 20 rounds. Color intensity indicates density. CorBin-FL’s errors cluster near the origin, aligning with Theorem 1.

Techniques such as gradient quantization have been considered in the literature mainly as a way to reduce the communication overhead in FL. Additionally, it has been observed that quantization may yield improved DP guarantees. In particular, a stochastic quantization method, called LDP-FL, was introduced in [12], which, in addition to reducing the communication overhead via one-bit quantization of each parameter update, guarantees parameter-level LDP (PLDP). Alternative mechanisms adding discrete noise to quantized updates to achieve DP have also been considered, such as the binomial mechanism in [15], [16].

In this work, we consider correlated stochastic quantization to achieve DP in FL. A key feature of our framework is that clients use shared common randomness to coordinate while maintaining privacy. To elaborate, at each communication round the server first generates a random pairing among the participating clients and, each assigned pair of clients then established a pairwise secret key through the Diffie-Hellman (DH) protocol [17], with the server acting as a blind relay for public DH values. The application of the DH protocol to construct pairwise secure communication among the clients has been previously explored in the literature, for instance the secure aggregation mechanism [3]. Using their shared secret key, each pair of clients securely exchange a compact PRG seed through the server, then both clients in this pair locally expand this seed into a shared random sequence, which is used to perform correlated stochastic quantization (Algorithm 2) of the model parameters. The quantizer is designed to achieve PLDP. Furthermore, the quantizer is unbiased and the quantization noises are correlated, such that they cancel each other in the aggregation phase, yielding small mean square error (MSE). Figure 1 visualizes a concrete example, where such reduction in MSE can be observed in FL over the MNIST dataset with PLDP constraints. We refer to the resulting privacy mechanism, which uses correlated binary stochastic quantization, as the CorBin-FL mechanism. We also introduce the AugCorBin-FL mechanism, in which a fraction of the clients pair up and

perform correlated quantization as in CorBin-FL, and the rest of the clients quantize their model weights individually. We show that the AugCorBin-FL mechanism achieves CDP and PLDP guarantees and quantify the corresponding privacy parameters and achievable MSE.

Achieving differential privacy with low MSE in federated learning through correlated mechanisms faces fundamental challenges. Gaussian-based approaches can theoretically achieve optimal MSE via maximal negative correlation [18], [19], yet they necessitate asymptotically infinite shared randomness per parameter [20]. In practice, implementations rely on finite pseudorandom seeds to generate correlated Gaussian noise, however the resulting discretization introduces additional vulnerabilities [14], and the gap between the finite seed approximation and true maximal negative correlation remains unqualified in these works. Conversely, while correlated quantization methods [21] offer practical communication efficiency, they remain fundamentally incompatible with formal differential privacy frameworks.

CorBin-FL addresses these limitations by introducing a novel correlated binary stochastic quantization framework that simultaneously satisfies PLDP and achieves near-optimal MSE using only finite shared randomness. Theoretically, our analysis assumes access to perfectly shared random bits, under which Theorem 1 provides a precise, finite bound on the MSE gap from optimality as a function of d . In practice, the shared random sequence can be generated from a compact PRG seed, which is the standard assumption in cryptographic implementations and suffices for all computational security guarantees. For scenarios requiring information theoretic security, the exact shared randomness can be exchanged directly, requiring $dm/2$ additional bits per client (still far below the overhead of Gaussian-based approaches). This separation between the theoretical model and practical implementation is made explicit in Remark 1. Experimental results on different datasets confirm that CorBin-FL outperforms standard non-correlated quantizers (LDP-FL) and traditional privacy-preserving mechanisms, such as Gaussian and Laplace noise injection, while maintaining a strict 1-bit-per-parameter quantization budget. Furthermore, our framework extends naturally to non-private settings, matching the performance of state-of-the-art methods [21] while providing the architectural flexibility to enable rigorous privacy guarantees as required. Our contributions are summarized as follows:

- To characterize a class of correlated stochastic quantizers optimizing the privacy-utility tradeoff (PUT) in distributed mean estimation. (Theorem 1).
- To introduce the CorBin-FL mechanism (Algorithm 3), and provide privacy and utility guarantees in terms of PLDP and MSE, together with a concrete analysis of communication overhead under both exact shared randomness and PRG-based implementations. (Theorem 2, Table 2)

- To introduce the AugCorBin-FL mechanism, and derive performance guarantees in terms of PLDP, CDP, and MSE. (Theorem 3)
- To provide empirical comparisons with the Gaussian mechanism, the Laplacian mechanism, and LDP-FL in various vision and natural language datasets, demonstrating PUT gains in CorBin-FL.

Notation: Sets are denoted by calligraphic letters such as $\mathcal{X}$. The set $\{1, 2, \dots, n\}$ is represented by $[n]$. Vectors and matrices are denoted by bold-face letters such as $\mathbf{x}$ and $\mathbf{H}$. The symbol $\prec$ denotes the lexicographic ordering on binary vectors. For $p \in \mathbb{N}$, we write $\|\cdot\|_p$ to denote the L_p -norm. Upper-case letters such as X represent random variables, and lower-case letters such as x represent their realizations. Similarly, random vectors and random matrices are denoted by upper-case letters such as $\mathbf{X}$ and $\mathbf{H}$, respectively.

II. PRELIMINARIES

A. PRIVACY MECHANISMS AND FEDERATED LEARNING

The setting comprises n clients, denoted by $\mathcal{C}_i, i \in [n]$, collaborating with a central server $\mathcal{S}$ to train a global model iteratively over multiple communication rounds. Each client $\mathcal{C}_i$ possesses a local dataset $\mathcal{D}_i$. At each communication round, the server sends the current global model to all clients, i.e., it sends the vector $\mathbf{w}_g = (w_{g,j})_{j \in [m]}$, where m is the number of model parameters. The i th client updates its local model based on $\mathcal{D}_i$ and computes the updated local model parameters $\mathbf{w}_i = (w_{i,j})_{j \in [m]}, i \in [n]$. For any given $j \in [m]$, let the bounded interval $[c_j - r_j, c_j + r_j]$ be chosen such that it contains the collection of updated local model parameters, $w_{i,j}, i \in [n]$. The variable c_j is called the *center* and the variable r_j the *radius*¹ corresponding to the j th model parameter. A key feature of our framework is that clients share a common vector of binary symmetric random variables $\mathbf{Z}^{d \times m}$, where d is the number of shared bits per model parameter. We will describe the method to share this common vector among the clients in the subsequent sections. Clients apply a (possibly stochastic) privacy mechanism $\mathcal{M} : \mathbb{R}^m \times \mathbb{R}^{d \times m} \rightarrow \mathbb{R}^m$, generating the obfuscated vectors $\bar{\mathbf{w}}_i = \mathcal{M}(\mathbf{w}_i, \mathbf{Z}^{d \times m})$, which are transmitted to the server. The server aggregates the obfuscated vectors, computing the updated parameters $\bar{\mathbf{w}}_g = \frac{1}{n} \sum_{i \in [n]} \bar{\mathbf{w}}_i$. The mechanism $\mathcal{M}$ should satisfy differential privacy constraints, as detailed in subsequent sections, while maintaining overall utility, in terms of the global model accuracy. In our theoretical derivations, the MSE between the aggregate of the obfuscated local updates and that of the unobfuscated updates is used as a surrogate utility measure. Our empirical evaluations show that the mechanism achieves improved accuracy over

widely used methods such as the Gaussian and Laplacian mechanisms.

B. LOCAL AND CENTRAL DIFFERENTIAL PRIVACY

Central Differential Privacy: For $\epsilon_c, \delta > 0$, a mechanism is said to achieve (ϵ_c, δ) -CDP if:

$$P(\bar{\mathbf{w}}_g \in \mathcal{T}) \leq e^{\epsilon_c} P(\bar{\mathbf{w}}'_g \in \mathcal{T}) + \delta, \quad (1)$$

$$\bar{\mathbf{w}}_g = \frac{1}{n} \sum_{i \in [n]} \bar{\mathbf{w}}_i, \quad \bar{\mathbf{w}}'_g = \frac{1}{n} \bar{\mathbf{w}}'_I + \frac{1}{n} \sum_{i \in [n] - I} \bar{\mathbf{w}}_i$$

where I is distributed uniformly over $[n]$, $\bar{\mathbf{w}}_i = \mathcal{M}(\mathbf{w}_i, \mathbf{Z}^{d \times m})$, $\mathbf{w}_i \in \mathbb{R}^m$, $\bar{\mathbf{w}}'_I = \mathcal{M}(\mathbf{w}'_I, \mathbf{Z}^{d \times m})$, $\mathbf{w}'_I \in \mathbb{R}^m$ such that $\|\mathbf{w}_I - \mathbf{w}'_I\|_2 \leq \Delta$, and $\mathcal{T} \subseteq \mathbb{R}^m$.

Parameter-Level Local Differential Privacy: Following [11], [12], for $\epsilon_p > 0$, a mechanism $\mathcal{M}$ is said to achieve ϵ_p -PLDP if:

$$\max_{j \in [m]} \frac{P(\bar{\mathbf{w}}_{i,j} \in \mathcal{T})}{P(\bar{\mathbf{w}}'_{i,j} \in \mathcal{T})} \leq e^{\epsilon_p}, \quad (2)$$

for all $i \in [n]$, where $\mathcal{T} \subseteq \mathbb{R}^m$, $\bar{\mathbf{w}}_i = \mathcal{M}(\mathbf{w}_i, \mathbf{Z}^{d \times m})$ and $\bar{\mathbf{w}}'_i = \mathcal{M}(\mathbf{w}'_i, \mathbf{Z}^{d \times m})$, and $\mathbf{w}_i, \mathbf{w}'_i \in \mathbb{R}^m$.

Parameter-Level Local Differential Privacy (PLDP) guarantees the privacy of each local model parameter if the uplink communication between client and server is compromised, consistent with the differential privacy definitions in [12], [22]–[24].

C. THE LDP-FL MECHANISM

In the subsequent sections, we build upon the LDP-FL mechanism [12] to introduce the CorBin-FL and AugCorBin-FL privacy mechanisms. For completeness, we provide a brief summary of the LDP-FL mechanism in this section. For a fixed $\epsilon_p > 0$, the mechanism uses a stochastic quantizer, denoted by $\text{LDPQ}(\cdot)$, to obfuscate each of the local model parameters. The $\text{LDPQ}(\cdot)$ quantizer is described in Algorithm 1. To provide an overview, given an input weight w , center c , and radius r , the quantizer produces a binary random variable U with

$$P(U = 1) = 1 - P(U = -1) = \frac{1}{2} + \frac{w - c}{2r\alpha(\epsilon_p)},$$

where $\alpha(\epsilon_p) = \frac{1+e^{\epsilon_p}}{1-e^{\epsilon_p}}$. Each client computes $\bar{w}_{i,j} = \text{LDPQ}(\epsilon_p, c_j, r_j, w_{i,j}), j \in [m]$ and sends it to the server.

Let us consider a scenario with no privacy requirements, i.e., $\epsilon_p \rightarrow \infty$. Then, $\alpha(\epsilon_p) \rightarrow 1$ and if $w_{i,j} > c_j$, then $P(U = 1) > P(U = -1)$, i.e., it is more likely that the client sends the update $\bar{w}_{i,j} = c_j + r_j$. Conversely, if $w_{i,j} < c_j$, then it is more likely that it sends the update $\bar{w}_{i,j} = c_j - r_j$. This can be interpreted as the client sending the direction of change, rather than the actual value of the updated model parameter.

As the privacy budget ϵ_p is decreased, the client update becomes less *truthful*, i.e., it becomes more likely that U points to the direction opposite to the true update direction. On average, the *misdirections* of different clients average out

¹The center c_j and radius r_j depend on the range of $w_{i,j}, i \in [n]$. In practice, the clients do not have access to each other's updated parameters. Consequently, the updated parameters are clipped by each client using a $\hat{c}_j$ and $\hat{r}_j$, which are shared by the server at each round to ensure that $w_{i,j} \in [\hat{c}_j - \hat{r}_j, \hat{c}_j + \hat{r}_j]$.

Algorithm 1 LDPQ: Binary-Output Stochastic Quantization**Procedure:** LDPQ(ϵ_p, c, r, w)**Inputs:** Privacy budget ϵ_p , center c , radius r , weight w

- 1: $\alpha(\epsilon_p) \leftarrow \frac{e^{\epsilon_p} + 1}{e^{\epsilon_p} - 1}$
- 2: $q \leftarrow \frac{1}{2} + \frac{w - c}{2r\alpha(\epsilon_p)}$
- 3: $U \leftarrow \text{BINRAND}(q)$ {generates binary U with $P_U(1) = q$ }
- 4: $Q(w) \leftarrow c + Ur\alpha(\epsilon_p)$
- 5: **return** $Q(w)$

since the mechanism produces an unbiased estimate, and the aggregate update on the server-side remains accurate. To see this, note that from Line 4 in Algorithm 1, we have $\mathbb{E}(U) = \frac{w_{i,j}}{2r_j\alpha(\epsilon_p)}$, so that $\mathbb{E}(\overline{W}_{i,j}) = w_{i,j}$. When the number of participating clients is sufficiently large, the unbiased property ensures that the aggregate of the obfuscated parameters closely approximates that of the original, unobfuscated inputs. The MSE of the parameter estimate is $\mathbb{E}((\overline{W}_{g,j} - w_{g,j})^2) \approx \frac{r_j^2 \alpha^2(\epsilon_p)}{n}$ [12], and vanishes as n becomes asymptotically large.

III. CORRELATED STOCHASTIC QUANTIZERS

In this section, we introduce a novel class of correlated stochastic quantizers. We first show that the LDPQ quantizer is the solution to a specific privacy-utility optimization. Then, we build on this to construct the maximally correlated distributed quantizers which are used in the CorBin-FL and AugCorBin-FL mechanisms in the subsequent sections.

A. OPTIMALITY OF THE LDPQ STOCHASTIC QUANTIZER

Given $c, r > 0$, a binary-output stochastic quantizer is a (stochastic) mapping $Q : [c - r, c + r] \rightarrow \{\gamma_1, \gamma_2\}$, where $\gamma_1, \gamma_2 \in \mathbb{R}$. We quantify privacy in terms of PLDP and utility in terms of the output bias and the MSE. To elaborate, for a given $\epsilon_p > 0$, we say the quantizer optimizes the PUT if the following conditions are satisfied:

- C1. **Unbiasedness:** $\mathbb{E}(Q(w)) = w$ for all $w \in [c - r, c + r]$,
- C2. ϵ_p -**PLDP:** $\frac{P(Q(w) = \overline{w})}{P(Q(w') = \overline{w})} \leq e^{\epsilon_p}$ for all $\overline{w} \in \{\gamma_1, \gamma_2\}$ and $w, w' \in [c - r, c + r]$.
- C3. **Minimal MSE:** $\mathbb{E}((Q(w) - w)^2)$ is minimized for all $w \in [c - r, c + r]$.

The following proposition shows that the LDPQ quantizer uniquely satisfies conditions C1-C3.

Proposition 1. *Let $c, r, \epsilon_p > 0$. The binary quantizer $Q(w) = \text{LDPQ}(\epsilon_p, c, r, w)$, $w \in [c - r, c + r]$ in Algorithm 1 is the unique quantizer satisfying conditions C1-C3.*

The proof is provided in Appendix A in the supplementary material.

B. CORRELATED STOCHASTIC QUANTIZATION

In the next step, we consider the distributed quantization of a pair of inputs w, w' when the pair of clients have access

Algorithm 2 CorBinQ: Correlated Stochastic Quantization**Procedure:** CORBINQ($\epsilon_p, c, r, w, w', d, \mathbf{Z}$)**Inputs:** Privacy budget ϵ_p , center c , radius r , weights w, w' , number of shared random bits d , shared random vector $\mathbf{Z}$

- 1: $\alpha(\epsilon_p) \leftarrow \frac{e^{\epsilon_p} + 1}{e^{\epsilon_p} - 1}$
- 2: $q_1 \leftarrow \frac{1}{2} + \frac{w - c}{2r\alpha(\epsilon_p)}$, $q_2 \leftarrow \frac{1}{2} + \frac{w' - c}{2r\alpha(\epsilon_p)}$
- 3: $\mathbf{T}_1 \leftarrow B_d(\lfloor 2^d q_1 \rfloor)$, $\mathbf{T}_2 \leftarrow B_d(\lfloor 2^d (1 - q_2) \rfloor)$
- 4: **if** $\mathbf{Z} \prec \mathbf{T}_1$ **then**
- 5: $Q_1(w, \mathbf{Z}) \leftarrow c + r\alpha(\epsilon_p)$
- 6: **else if** $\mathbf{T}_1 \prec \mathbf{Z}$ **then**
- 7: $Q_1(w, \mathbf{Z}) \leftarrow c - r\alpha(\epsilon_p)$
- 8: **else**
- 9: $U \leftarrow \text{BINRAND}(2^d q_1 - \lfloor 2^d q_1 \rfloor)$
- 10: $Q_1(w, \mathbf{Z}) \leftarrow c + Ur\alpha(\epsilon_p)$
- 11: **end if**
- 12: **if** $\mathbf{Z} \prec \mathbf{T}_2$ **then**
- 13: $Q_2(w', \mathbf{Z}) \leftarrow c - r\alpha(\epsilon_p)$
- 14: **else if** $\mathbf{T}_1 \prec \mathbf{Z}$ **then**
- 15: $Q_2(w', \mathbf{Z}) \leftarrow c + r\alpha(\epsilon_p)$
- 16: **else**
- 17: $U' \leftarrow \text{BINRAND}(2^d (1 - q_2) - \lfloor 2^d (1 - q_2) \rfloor)$
- 18: $Q_2(w', \mathbf{Z}) \leftarrow c - U'r\alpha(\epsilon_p)$
- 19: **end if**
- 20: **return** $Q_1(w, \mathbf{Z}), Q_2(w', \mathbf{Z})$

to a shared sequence of common random bits. To elaborate, let us assume that the clients have access to a sequence of binary symmetric random variables $\mathbf{Z} = (Z_1, Z_2, \dots, Z_d)$ for a fixed $d \in \mathbb{N}$. We wish to design a pair of stochastic quantizers $Q_i : [c - r, c + r] \times \{0, 1\}^d \rightarrow \{\gamma_1, \gamma_2\}$, $i \in \{1, 2\}$, satisfying the following conditions:

- C4. **Unbiased Output:** $\mathbb{E}(Q_1(w), \mathbf{Z}) = w$, $\mathbb{E}(Q_2(w'), \mathbf{Z}) = w'$ for all $w, w' \in [c - r, c + r]$.
- C5. ϵ_p -**PLDP:** $\frac{P(Q_i(w, \mathbf{Z}) = \overline{w})}{P(Q_i(w', \mathbf{Z}) = \overline{w})} \leq e^{\epsilon_p}$ for all $\overline{w} \in \{\gamma_1, \gamma_2\}$, $w, w' \in [c - r, c + r]$, and $i \in \{1, 2\}$.
- C6. **Minimal MSE:** $\mathbb{E}((Q_1(w, \mathbf{Z}) + Q_2(w', \mathbf{Z}) - w - w')^2)$ is minimized for all $w, w' \in [c - r, c + r]$.

The pair of stochastic quantizers used in CorBin-FL, denoted by CORBINQ are described in Algorithm 2. In the sequel, we show that the pair satisfies conditions C4-C5 for all $d \in \mathbb{N}$, and condition C6 asymptotically as $d \rightarrow \infty$. To provide an overview, the CORBINQ quantizers described in Algorithm 2 takes parameters $\epsilon_p, r, c > 0$, weights $w, w' \in [c - r, c + r]$, and a shared sequence of independent binary symmetric random variables $\mathbf{Z}$ as input. It then constructs a pair of *threshold vectors* $\mathbf{T}_1$ and $\mathbf{T}_2$ as described in Line 3 of the algorithm, where $B_d(\cdot)$ denotes the d -bit decimal to binary operator. The quantization process involves comparing the shared random sequence $\mathbf{Z}$ with these threshold vectors using lexicographic ordering ($\prec$). To elaborate, as described in Lines 4-11 of the algorithm, the first quantizer, Q_1 , outputs $c + r\alpha(\epsilon_p)$ if $\mathbf{Z} \prec \mathbf{T}_1$ and $c - r\alpha(\epsilon_p)$ if $\mathbf{T}_1 \prec \mathbf{Z}$. Ties are resolved using a locally

generated binary random variable U with bias $2^d q_1 - \lfloor 2^d q_1 \rfloor$, as described in Lines 9-10. On the other hand, as described in Lines 12-19, the second quantizer, Q_2 , outputs $c - r\alpha(\epsilon_p)$ if $\mathbf{Z} \prec \mathbf{T}_2$ and $c + r\alpha(\epsilon_p)$ if $\mathbf{T}_2 \prec \mathbf{Z}$, with ties resolved by U' as detailed in Lines 17-18. The design of $\mathbf{T}_i$ and the tie-breaking rule ensures that the marginal distribution of the output of each Q_i matches that of the stochastic quantizer used in LDP-FL. Thus, satisfying the unbiasedness and ϵ_p PLDP conditions. Furthermore, Q_1 and Q_2 produce outputs based on reverse lexicographic ordering relative to each other. This guarantees maximum negative correlation between the quantizers, which minimizes the resulting MSE. The following theorem shows that i) the pair of quantizers (Q_1, Q_2) in Algorithm 2 satisfy conditions C4-C5 for all $d \in \mathbb{N}$, ii) they optimize the privacy-utility tradeoff described by conditions C4-C6 asymptotically as $d \rightarrow \infty$, and iii) provides an upper-bound on the MSE for $d \in \mathbb{N}$. The proof is provided in Appendix B in the supplementary material.

Theorem 1. *Let $d \in \mathbb{N}$ and $r, c, \epsilon_p > 0$. The pair of correlated stochastic quantizers (Q_1, Q_2) in Algorithm 2 satisfy conditions C4-C5. Furthermore, let $\mathcal{Q}$ consist of all pairs (Q'_1, Q'_2) satisfying conditions C4-C5, and let the MSE of (Q'_1, Q'_2) for inputs (w, w') be defined as:*

$$m(Q'_1, Q'_2, w, w') = \mathbb{E}((Q'_1(w, \mathbf{Z}) + Q'_2(w', \mathbf{Z}) - w - w')^2),$$

for all (Q'_1, Q'_2) $\in \mathcal{Q}$. Then, the MSE associated with (Q_1, Q_2) is bounded from above as follows:

$$m(Q_1, Q_2, w, w') \leq \min_{(Q'_1, Q'_2) \in \mathcal{Q}} m(Q'_1, Q'_2, w, w') + \frac{r^2 \alpha^2(\epsilon_p)}{2^{d-6}}, \quad (3)$$

for all $w, w' \in [r - c, r + c]$. Particularly, $m(Q_1, Q_2, w, w')$ converges uniformly to the minimum MSE over $\mathcal{Q}$ as $d \rightarrow \infty$.

IV. CORBIN-FL AND AUGCORBIN-FL MECHANISMS

A. THE CORBIN-FL MECHANISM

The mechanism is shown in Algorithm 3. The server first generates a random pairing $\{(i, p_i)\}_{i \in [n]}$ on $[n]$, chosen uniformly on all possible pairings. The server then broadcasts the global model weights $\mathbf{w}_g$, center and radius vectors $\mathbf{c}$ and $\mathbf{r}$, the pairing, and privacy budget ϵ_p to all clients (Step 1, Lines 1–2). For each assigned pair $(\mathcal{C}_i, \mathcal{C}_{p_i})$, a pairwise shared secret key K_{i, p_i} is established via the DH protocol (Step 2, Lines 3–5), with the server acting and communication facilitator. The server observes only the public values and cannot compute K_{i, p_i} . Since each client perform DH only with its assigned partner, this incurs $\mathcal{O}(k)$ communication per client, where k is the DH key size [17]. Each client then computes its local model update $\mathbf{w}_i$ from its local dataset $\mathcal{D}_i$ (Step 3, Line 6).

To assign lead (ℓ) and follow (f) roles within each pair (Step 4, Lines 8–14), both clients generate independent binary random variables Y_i and Y_{p_i} , exchange them via their encrypted channel facilitated by server, and assign roles based on agreement or disagreement of the values.

Algorithm 3 CorBin-FL Mechanism

Procedure: CORBINFL($n, m, \epsilon_p, \mathbf{c}, \mathbf{r}, d, \mathbf{w}_g, (\mathcal{D}_i)_{i \in [n]}, \mathcal{S}$)
Inputs: Number of clients n , number of model parameters m , privacy budget ϵ_p , center vector $\mathbf{c}$, radius vector $\mathbf{r}$, number of shared bits per parameter d , global model parameters $\mathbf{w}_g$, local datasets $\mathcal{D}_i, i \in [n]$, server $\mathcal{S}$

Step 1: Server generates pairing

1: $\{(i, p_i)\}_{i \in [n]} \leftarrow \text{GENERATERANDOMPAIRING}(n)$
 {Executed by $\mathcal{S}$ }

2: $\mathcal{S}$ broadcasts $\mathbf{w}_g, \mathbf{c}, \mathbf{r}, \{p_i\}_{i \in [n]}, \epsilon_p$ to all clients

Step 2: Pairwise DH key establishment

3: **for** each pair $(\mathcal{C}_i, \mathcal{C}_{p_i}), i < p_i$ **do**

4: $K_{i, p_i} \leftarrow \text{DHE}(\mathcal{C}_i, \mathcal{C}_{p_i}, \mathcal{S})$ { $\mathcal{S}$ relays public DH values only}

5: **end for**

Step 3: Local model update

6: Each $\mathcal{C}_i$ computes update $\mathbf{w}_i$ from $\mathcal{D}_i$

Step 4: lead/follow role assignment

7: **for** each pair $(\mathcal{C}_i, \mathcal{C}_{p_i})$ **do**

8: $Y_i \leftarrow \text{BINRAND}(1/2), Y_{p_i} \leftarrow \text{BINRAND}(1/2)$

9: $\text{ENCRYPTRELAY}(\mathcal{C}_i, \mathcal{C}_{p_i}, (Y_i, Y_{p_i}), K_{i, p_i}, \mathcal{S})$

10: **if** $Y_i = Y_{p_i}$ **then**

11: $\ell \leftarrow \mathcal{C}_{\min(i, p_i)}, f \leftarrow \mathcal{C}_{\max(i, p_i)}$

12: **else**

13: $\ell \leftarrow \mathcal{C}_{\max(i, p_i)}, f \leftarrow \mathcal{C}_{\min(i, p_i)}$

14: **end if**

Step 5: Shared randomness generation

15: **[Practical]:** ℓ generates 32-bit seed s , computes $\mathbf{Z}_{i, j} \leftarrow \text{PRG}(s, j)$ for $j \in [m]$

16: $\text{ENCRYPTRELAY}(\ell, f, s, K_{i, p_i}, \mathcal{S})$ {Only 32 bits relayed through $\mathcal{S}$ }

17: f computes $\mathbf{Z}_{i, j} \leftarrow \text{PRG}(s, j)$ for $j \in [m]$

18: **[Exact]:** ℓ generates $\mathbf{Z}_{i, j} \leftarrow [\text{BINRAND}(1/2)]^d$ for $j \in [m]$ and sends via $\text{ENCRYPTRELAY}(\ell, f, \{\mathbf{Z}_{i, j}\}_{j=1}^m, K_{i, p_i}, \mathcal{S})$

Step 6: Correlated quantization

19: **for** $j \in [m]$ **do**

20: $(\bar{\mathbf{W}}_{i, j}, \bar{\mathbf{W}}_{p_i, j})$

$\text{CORBINQ}(\epsilon_p, c_j, r_j, w_{i, j}, w_{p_i, j}, d, \mathbf{Z}_{i, j})$

21: **end for**

22: $\mathcal{C}_i, \mathcal{C}_{p_i}$ upload $\bar{\mathbf{W}}_i, \bar{\mathbf{W}}_{p_i}$ to $\mathcal{S}$

23: **end for**

24: $\bar{\mathbf{W}}_g \leftarrow \frac{1}{n} \sum_{i \in [n]} \bar{\mathbf{W}}_i$

this ensures the role assignment is symmetric and unpredictable to server. The lead client then generates shared randomness for the correlated quantization (Step 5, Lines 15–18). Under the practical PRG-based implementation, the lead generates 32-bit seed (s), encrypts it using K_{i, p_i} , and send it to the follow client through server. Both clients locally expand the seed into the shared sequence $\{\mathbf{Z}_{i, j}\}_{j=1}^m$ using a cryptographically secure PRG. The server cannot

decrypt the seed and therefore learns nothing about $\mathbf{Z}$. Under the exact implementation, the full random sequence is transmitted directly, at higher but bounded communication cost (see Remark 1). Each pair applies CORBINQ to generate correlated quantized updates $(\bar{\mathbf{W}}_i, \bar{\mathbf{W}}_{p_i})$ (Step 6, Lines 19–23), which are uploaded to the server. The server aggregates all updates: $\bar{\mathbf{W}}_g \leftarrow \frac{1}{n} \sum_{i \in [n]} \bar{\mathbf{W}}_i$ (Line 24). Note that if $d = 0$, the CorBin-FL mechanism reduces to LDP-FL.

From Theorem 1 it follows that CorBin-FL achieves ϵ_p -PLDP. The following theorem states the utility guarantees of CorBin-FL, in terms of unbiasedness and achievable MSE on the server-side.

Theorem 2. *The CorBin-FL mechanism is unbiased, i.e., $\mathbb{E}(\bar{\mathbf{W}}_g) = \frac{1}{n} \sum_{i \in [n]} \mathbf{w}_i$, where $\mathbf{w}_i, i \in [n]$ are the local client updates, and $\bar{\mathbf{W}}_g$ is the average of the obfuscated updates at the server. Furthermore, the MSE is bounded by:*

$$\begin{aligned} & \lim_{d \rightarrow \infty} \mathbb{E}((\bar{\mathbf{W}}_{g,j} - \frac{1}{n} \sum_{i \in [n]} w_{i,j})^2) \\ & \leq \frac{r_j^2}{2n} \left((\sqrt{2}-1)\alpha(\epsilon_p)+1 \right) \left((\sqrt{2}+1)\alpha(\epsilon_p)-1 \right), \quad j \in [m] \end{aligned}$$

The proof is provided in Appendix C. As shown in [12], the mean square error under LDP-FL is close to $\frac{r^2 \alpha^2(\epsilon_p)}{n}$ for large n . Consequently, for small values of ϵ_p , where $\alpha(\epsilon_p) \gg 1$ and large values of n , the CorBin-FL mechanism improves the mean-square error by at least a factor of two compared to LDP-FL.

Remark 1 (Exact and PRG-based shared randomness). *The theoretical analysis in Theorem 1 and Theorem 2 assumes that the shared random sequence $\{\mathbf{Z}_{i,j}\}_{j=1}^m$ consists of perfectly independent binary symmetric random variables. Under the **exact** implementation (Algorithm 3, Line 18), this is satisfied exactly at a cost of $dm/2$ additional bits per client. Under the **PRG-based** implementation (Algorithm 3, Line 15), the sequence is generated from a 32-bit seed using a cryptographically secure PRG, reducing the communication cost to 16 additional bits per client. The PRG-based sequence is computationally indistinguishable from truly random under standard cryptographic assumptions, and all practical security guarantees are preserved. For applications requiring information theoretic security, the exact implementation should be used.*

B. THE AUGMENTED CORBIN-FL MECHANISM

So far, we have shown that CorBin-FL is unbiased, achieves ϵ_p -PLDP, and improves upon LDP-FL in terms of MSE. In addition to PLDP, the CorBin-FL mechanism can be modified to achieve CDP. To elaborate, let $\gamma \in [0, 1]$ and consider a mechanism where, at each round, γ fraction of the clients use the LDP-FL mechanism to obfuscate their corresponding weights, and $(1-\gamma)$ fraction use the CorBin-FL mechanism. We call the resulting mechanism, which is

a hybrid mechanism between LDP-FL and CorBin-FL, the AugCorBin-FL mechanism. The following theorem prides the UCDP guarantees and an upper-bound on the mean square error resulting from the AugCorBin-FL mechanism.

Theorem 3. *Let $\epsilon_p, \delta > 0$, and $\gamma \in (0, 1]$. Let us define $r = \max_{j \in [m]} r_j$ and assume that n is large enough that*

$$(n\gamma - 1) \left(\frac{1}{4} - \frac{1}{4\alpha^2(\epsilon_p)} \right) \geq \max(23 \log \frac{m}{\delta}, 2r\alpha(\epsilon_p)).$$

Then, the AugCorBin-FL mechanism achieves (ϵ_c, δ) -CDP and ϵ_p -PLDP, where

$$\frac{\epsilon_c}{r\alpha(\epsilon_p)} = \sqrt{\frac{8m \log \frac{1.25}{\delta}}{(n\gamma - 1)e_p}} + \frac{8(\log \frac{1.25}{\delta} + \log \frac{20m}{\delta} \log \frac{10}{\delta})}{3(n\gamma - 1)} \quad (4)$$

$$\begin{aligned} & + \frac{4b_p \sqrt{2m} (1.75 + \frac{3.75}{\alpha^2(\epsilon_p)}) \sqrt{\log \frac{10}{\delta}}}{(n\gamma - 1)(1 - \frac{\delta}{10})e_p}, \\ & e_p = (1 + \frac{1}{\alpha^2(\epsilon_p)}), \quad b_p = \frac{1}{3}e_p + \frac{1}{\alpha(\epsilon_p)}. \end{aligned}$$

Furthermore, the mean square error is bounded by:

$$\begin{aligned} & \lim_{d \rightarrow \infty} \mathbb{E}((\bar{\mathbf{W}}_{g,j} - \frac{1}{n} \sum_{i \in [n]} w_{i,j})^2) \leq \frac{\gamma r_j^2 \alpha^2(\epsilon_p)}{n} + \\ & \frac{(1-\gamma)}{2n} r_j^2 \left((\sqrt{2}-1)\alpha(\epsilon_p)+1 \right) \left((\sqrt{2}+1)\alpha(\epsilon_p)-1 \right), \end{aligned}$$

for all $j \in [m]$.

The proof is provided in Appendix D.

1) Robustness to Client Dropout

The CorBin-FL mechanism is robust to client dropouts. The reason is that if a member of a pair of clients drops out, the quantization performed by the other client is equivalent to LDPQ. In fact, if a lower-bound to the dropout probability of each individual client is known, and if the clients dropout independently of each other, then one can derive additional central differential privacy guarantees for the resulting mechanism. This is made precise in the following theorem.

Theorem 4. *Let $\epsilon_p, \delta, p, t > 0$, and $\gamma \in (0, 1]$. Assume that clients drop out independently of each other with dropout probability p in each communication round. Furthermore, let us define $r = \max_{j \in [m]} r_j$ and assume that n is large enough and there exists $\delta_1 < \delta$ such that*

$$\begin{aligned} & (n\gamma - 1) \left(\frac{1}{4} - \frac{1}{4\alpha^2(\epsilon_p)} \right) \geq \max(23 \log \frac{m}{\delta_1}, 2r\alpha(\epsilon_p)), \\ & \gamma = p(1-p) - \sqrt{\frac{p(1-p)(1-2p(1-p))}{4(\delta - \delta_1)n}}. \end{aligned}$$

Then, the CorBin-FL mechanism achieves ϵ_p -PLDP and (ϵ_c, δ) -CDP, where, ϵ_c is given by Equation (4). Furthermore,

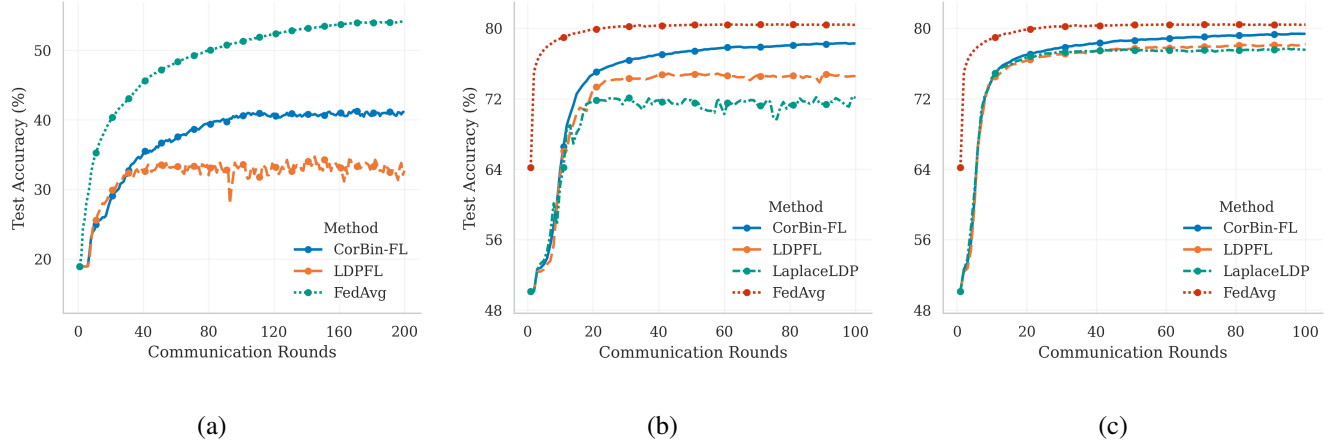


FIGURE 2. Experimental results for (a) test accuracy on Shakespeare with $\epsilon_p = 1$, (b) test accuracy on Sentiment140 with $\epsilon_p = 1$, and (c) test accuracy on Sentiment140 with $\epsilon_p = 3$.

the mean square error is bounded by:

$$\lim_{d \rightarrow \infty} \mathbb{E}((\bar{W}_{g,j} - \frac{1}{N} \sum_{i: F_i=1} w_{i,j})^2) \leq \frac{\gamma' r^2 \alpha^2(\epsilon_p)}{n\theta} + 8r^2 \alpha^2(\epsilon_p) \delta_1 + \frac{(1-\gamma)}{2n\theta} r^2 \left((\sqrt{2}-1)\alpha(\epsilon_p) + 1 \right) \left((\sqrt{2}+1)\alpha(\epsilon_p) - 1 \right).$$

for all $j \in [m]$, where F_i is the indicator that the i th client does not drop out, $N = \sum_{i \in [n]} F_i$, and

$$\gamma' = p(1-p) + \sqrt{\frac{p(1-p)(1-2p(1-p))}{4(\delta - \delta_1)n}},$$

$$\theta = p(1-p) - \sqrt{\frac{p(1-p)}{(\delta - \delta_1)n}}.$$

The proof is provided in Appendix E.

Communication Cost The server broadcasts the global model to all clients, which requires $32m$ bits of downlink communication, assuming 32-bit floating-point representation. The server then announce the random pairing $\{(i, p_i)\}_{i \in [n]}$ to all clients. For each assigned pair (C_i, C_{p_i}) , a pairwise secret key K_{i,p_i} is established via the DH protocol [17], with the server acting as communication facilitator. This incurs $2k$ bits of uplink communication per client and is negligible for large models since $2k \ll m$ for standard DH key size (e.g., $k = 2048$ bits) and typical model sizes (e.g. $m \geq 10^5$).

The lead client in each pair generates a 32-bit PRG seed, encrypts it using K_{i,p_i} , and uploads the encrypted seed to the server which sends it to the follow client. Both clients locally expand the seed into the shared random sequence $\{Z_{i,j}\}_{j=1}^m$ using a cryptographically secure PRG, requiring no further communication. Each client then applies CORBINQ and uploads its 1-bit quantized model update, requiring m bits of uplink communication. The total uplink cost per client per round is therefore $m + 2k + 16$ bits under the PRG-based implementation, where 16 bits accounts for half of 32-bit seed per pair or clients. For large models where $m \gg k$, this reduces effectively to m bits (identical to

independent quantization). Table 2 summarize the per-client uplink communication cost across methods, showing that CorBin-FL achieves MSE gains with negligible additional communication cost.

TABLE 2. Per-client uplink communication cost per round.

METHOD	UPLINK BITS	$m=10^6, d=5$
GAUSSIAN / LAPLACE	$32m$	32×10^6
LDP-FL	m	10^6
CORBIN-FL (RAW)	$m + dm/2$	3.5×10^6
CORBIN-FL (PRG)	$m + 2k + 16$	$\approx 10^6$

Computation Cost The computation cost of the DH key exchange protocol is $\mathcal{O}(k^2 \log k)$ per client, since each client performs single DH exchange with its assigned partner (constant in n). For generating dm binary symmetric random variables, the cost is $\mathcal{O}(dm)$, and the encryption and computational complexity of CorBinQ in Algorithm 2 consisting of a lexicographic binary vector comparison and the generation of a binary random variable is $\mathcal{O}(m)$ per client. In total the computation complexity is $\mathcal{O}(dm + k^2 \log k)$.

V. EMPIRICAL ANALYSIS

We empirically evaluate the performance of the proposed privacy mechanisms over a series of experiments as outlined in the sequel. To provide an overview, our objective is to demonstrate that i) under a fixed privacy budget and number of communication rounds, the proposed mechanisms outperform the LDP-FL, and Gaussian and Laplace Mechanisms in terms of overall model accuracy (Table 4), ii) the accuracy gains increase as the number of shared random bits per model parameter is increased, however, the gains saturate for relatively small values of d (e.g., $d = 5$) as predicted in Theorem 1 (Figure 3(a)), iii) the proposed methods are robust to client dropouts, and the model accuracy does not fall significantly even for large dropout probabilities, e.g.,

TABLE 1. Maximum test accuracies (%) for different epsilon on MNIST and CIFAR10 with 50 clients for 100 rounds.

DATASET	METHOD	$\epsilon_p = 0.1$	$\epsilon_p = 0.3$	$\epsilon_p = 1$	$\epsilon_p = 3$
MNIST	LDPFL	26.93 $\pm$ 1.51	88.27 $\pm$ 0.23	87.22 $\pm$ 0.55	97.02 $\pm$ 0.02
	LAPLACE	25.63 $\pm$ 1.97	78.01 $\pm$ 0.73	80.28 $\pm$ 0.35	95.98 $\pm$ 0.05
	CORBIN-FL	93.34 $\pm$ 0.30	97.00 $\pm$ 0.11	96.08 $\pm$ 0.04	97.72 $\pm$ 0.05
	FEDAVG	98.43 $\pm$ 0.04	98.43 $\pm$ 0.04	98.43 $\pm$ 0.04	98.43 $\pm$ 0.04
		$\epsilon_p = 1$	$\epsilon_p = 3$	$\epsilon_p = 5$	$\epsilon_p = 7$
CIFAR10	LDPFL	44.23 $\pm$ 0.29	65.69 $\pm$ 0.73	68.93 $\pm$ 0.51	73.12 $\pm$ 0.38
	LAPLACE	27.9 $\pm$ 3.44	63.64 $\pm$ 2.31	67.43 $\pm$ 0.30	71.38 $\pm$ 0.31
	CORBIN-FL	59.51 $\pm$ 0.24	68.84 $\pm$ 0.62	72.69 $\pm$ 0.48	73.41 $\pm$ 0.52
	FEDAVG	78.21 $\pm$ 0.64	78.21 $\pm$ 0.64	78.21 $\pm$ 0.64	78.21 $\pm$ 0.64

for $p = 0.5$ (Figure 3(b)), and iv) the proposed methods outperform the LDP-FL mechanism for different numbers of clients ranging from 50 to 500 (Figure 3(c)). A detailed description of the experimental setup is provided in this section. Further empirical evaluations and ablation studies are included in the technical appendix.

Experiments are conducted over MNIST [25], CIFAR10 [26], FEMNIST, Shakespeare and Sentiment140 datasets from the LEAF benchmark [27]. These span image classification (MNIST, CIFAR10, FEMNIST), next-character prediction (Shakespeare), and sentiment analysis (Sentiment140) tasks. For FEMNIST, we additionally evaluate a non-IID scenario where data is distributed based on the writers of the characters, reflecting a more realistic federated setting. A detailed description of the experimental setup is provided in this section.

Models: We use a ResNet18 for CIFAR10, a VGG-style network for MNIST, a CNN for FEMNIST, and LSTM-based models for both Shakespeare and Sentiment140. The model architectures are detailed in the Appendix H.

Hyperparameters: The number of shared random bits per parameter is $d = 5$. All experiments use the exact shared randomness implementation, corresponding to $dm/2$ additional uplink bits per client per round. Results under the PRG-based implementation are expected to be identical in accuracy, as the PRG sequence is computationally indistinguishable from the exact random sequence (Remark 1). The AugCorBin-FL parameter is $\gamma = 0.2$ across all experiments unless specified otherwise. The center and radius vectors are updated each round according to layer-wise parameter ranges from the previous round. We implement checkpointing with a patience of five epochs, reloading the weights if accuracy decreases over five epochs.

Smoothed Update Rule: We include a global learning rate hyperparameter λ . At each communication round, let $\mathbf{w}_{g,init}$ be the initial global parameter vector, and $\mathbf{w}_{g,final}$ the updated parameter vector at the end of the round. We update the model as $\mathbf{w}_g = (1 - \lambda)\mathbf{w}_{g,init} + \lambda\mathbf{w}_{g,final}$. For each experiment, and for all privacy mechanisms, we perform a grid search over $\lambda \in \{0.1, 0.2, \dots, 1\}$.

A. IMAGE CLASSIFICATION TASKS

We evaluate MNIST, CIFAR10, and FEMNIST in a setting with 50 clients over 100 communication rounds.

MNIST Experiments: CorBin-FL consistently outperforms LDP-FL and the Laplace mechanism across privacy budgets (Table 1). PCA analysis of quantization errors (Figure 1) confirms reduced MSE, aligning with Theorem 1.

CIFAR10 Experiments: Table 1 shows that CorBin-FL achieves higher accuracy across various PLDP budgets and consistently outperforms LDP-FL and the Laplace mechanism. Table 4 provides a more comprehensive comparison for $\epsilon_p = 5$, where we have compared the Gaussian, Laplace, LDP-FL, CorBin-FL, and AugCorBin-FL mechanisms, for a model trained over 300 rounds of communication. CorBin-FL achieves the best accuracy among private methods. To provide additional benchmarks, we have also provided the accuracy for CQ [21] and FedAvg.

TABLE 3. Test accuracies on FEMNIST with 50 clients for 100 rounds.

DATASET	METHOD	$\epsilon_p = 1$	$\epsilon_p = 5$
FEMNIST	LDPFL	79.98 $\pm$ 0.69	84.96 $\pm$ 0.46
	LAPLACE	80.75 $\pm$ 0.04	84.20 $\pm$ 0.06
	CORBIN-FL	82.82 $\pm$ 0.70	86.08 $\pm$ 0.44
	FEDAVG	87.24 $\pm$ 0.39	87.24 $\pm$ 0.39

FEMNIST Experiments: IID FL experiments show accuracy gains of CorBin-FL (Table 3). For non-IID results and their consistency across different dataset partitions, please refer to Figure 4 in the technical appendix.

B. NATURAL LANGUAGE TASKS

We perform experiments over the Shakespeare dataset with 150 clients (Figure 2a), and the Sentiment140 dataset with 50 clients. As shown in Figure 2(b,c), both experiments show accuracy gains over existing privacy mechanisms.

C. ROBUSTNESS, SCALABILITY, AND NUMBER OF SHARED BITS

Impact of Shared Random Bits: Figure 3a implements CorBin-FL with different number of shared random bits per model parameter $d \in \{3, 5, 7\}$. It can be observed that the accuracy gains saturate at around $d = 5$ bits.

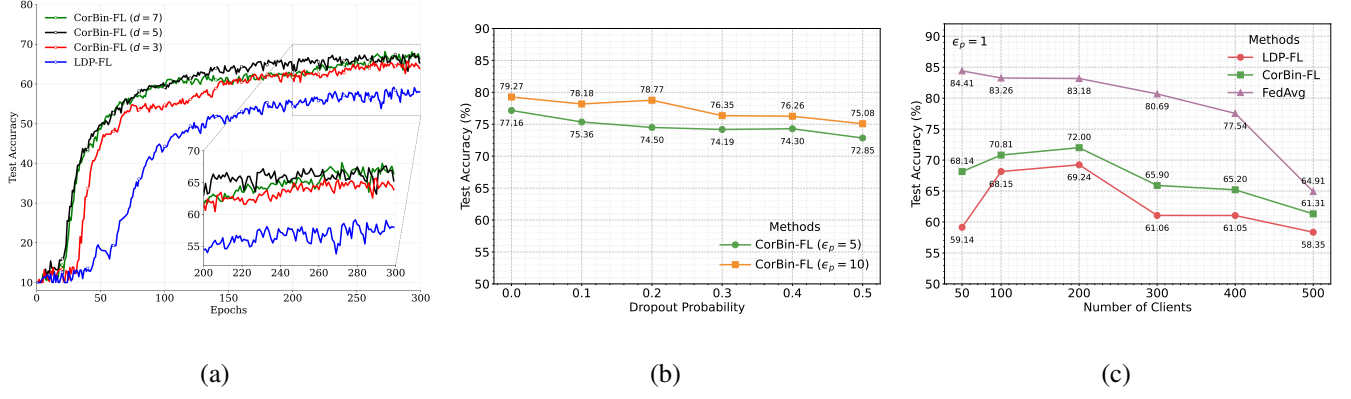


FIGURE 3. Experimental results for (a) number of shared common random bits, (b) dropout probability and (c) number of clients.

TABLE 4. Comparison of various privacy mechanisms and learning methods for CIFAR10 with 50 clients and 300 rounds.

METHOD	TEST ACCURACY (%)	ϵ_p
GAUSSIAN	57.01	5
LAPLACE	73.37	5
LDP-FL	74.20	5
AUGCORBIN-FL	75.60	5
CORBIN-FL	77.16	5
CQ	81.21	—
FEDAVG	84.41	—

System Robustness and Scalability: Figure 3b shows CorBin-FL's performance remains stable with up to 50% client dropout probability. Scalability experiments in Figure 3c demonstrate consistent performance when scaling from 50 to 500 clients, maintaining a stable gap with the non-private FedAvg baseline. It can be observed that the FedAvg accuracy decreases monotonically as a function of number of clients, whereas there is sweetspot for the number of clients in private mechanisms. The justification is that as the number of clients increases while the total dataset size remains fixed, the local data size per client decreases, reducing the statistical quality of each client's local update and the convergence rate of FedAvg. For private mechanisms, there is a tradeoff between the MSE of the obfuscation mechanism reducing with number of clients which yields accuracy gains versus the loss of accuracy due to reduced batch size for each client.

VI. CONCLUSION AND FUTURE WORKS

The CorBin-FL and AugCorBin-FL mechanisms were introduced for differentially private federated learning. These approaches, which are based on correlated binary stochastic quantization, were shown to achieve various notions of differential privacy. Theoretical guarantees for privacy parameters and mean squared error were derived. The proposed privacy mechanisms consistently outperformed LDP-FL, Gaussian, and Laplace mechanisms across multiple experiments. In future works, we will investigate the extension of the

proposed methods to correlated quantizers with larger output alphabets, and correlated quantization among collections of more than two clients.

Appendix A PROOF OF PROPOSITION 1

For $w \in [c - r, c + r]$, let $p_w = P(Q(w) = \gamma_1)$. From condition C1, we have:

$$p_w \gamma_1 + (1 - p_w) \gamma_2 = w \Rightarrow p_w = \frac{\gamma_2 - w}{\gamma_2 - \gamma_1}. \quad (5)$$

From condition C2, for $w, w' \in [c - r, c + r]$ and $\bar{w} = \gamma_1$, we have:

$$\frac{P(Q(w) = \bar{w})}{P(Q(w') = \bar{w})} \leq e^{\epsilon_p} \Rightarrow \frac{\frac{\gamma_2 - w}{\gamma_2 - \gamma_1}}{\frac{\gamma_2 - w'}{\gamma_2 - \gamma_1}} = \frac{\gamma_2 - w}{\gamma_2 - w'} \leq e^{\epsilon_p}.$$

To ensure that the condition holds for all $w, w' \in [c - r, c + r]$, it suffices to ensure the condition holds when the numerator is maximized and the denominator is minimized, i.e., for $w = c - r$ and $w' = c + r$. We have:

$$\frac{\gamma_2 - c + r}{\gamma_2 - c - r} \leq e^{\epsilon_p} \Rightarrow \gamma_2 \geq c + r \left(\frac{e^{\epsilon_p} + 1}{e^{\epsilon_p} - 1} \right) = c + r\alpha(\epsilon_p). \quad (6)$$

Similarly, by taking $\bar{w} = \gamma_2$, we have:

$$\gamma_1 \leq c - r \left(\frac{e^{\epsilon_p} + 1}{e^{\epsilon_p} - 1} \right) = c - r\alpha(\epsilon_p). \quad (7)$$

Note that

$$\begin{aligned} \mathbb{E}((Q(w) - w)^2) &= \mathbb{E}(Q^2(w)) - 2w\mathbb{E}(Q(w)) + w^2 \\ &= \mathbb{E}(Q^2(w)) - w^2, \end{aligned}$$

where we have used the unbiasedness property in the last equality. So, Condition C3 requires the minimization of the following:

$$\begin{aligned} \mathbb{E}(Q^2(w)) &= p_w \gamma_1^2 + (1 - p_w) \gamma_2^2 \\ &= \left(\frac{\gamma_2 - w}{\gamma_2 - \gamma_1} \right) \gamma_1^2 + \left(\frac{w - \gamma_1}{\gamma_2 - \gamma_1} \right) \gamma_2^2 \\ &= \frac{-\gamma_1 \gamma_2 (\gamma_2 - \gamma_1) + (\gamma_2 - \gamma_1)(\gamma_2 + \gamma_1)w}{\gamma_2 - \gamma_1} \\ &= -\gamma_1 \gamma_2 + (\gamma_1 + \gamma_2)w. \end{aligned}$$

Note that since $w \in [c - r, c + r] \subseteq [\gamma_1, \gamma_2]$ from Equations (6) and (7), the derivative of the above term is positive with respect to γ_2 for all values of γ_1 , and it is negative with respect to γ_1 for all values of γ_2 . Thus, from the constraints of Equations (6) and (7), we get $\gamma_1 = c - r\alpha(\epsilon_p)$ and $\gamma_2 = c + r\alpha(\epsilon_p)$. It remains to show that $P(Q(w) = c - r\alpha(\epsilon_p)) = \frac{1}{2} - \frac{w-c}{2r\alpha(\epsilon_p)}$, which follows from Equation (5):

$$p_w = \frac{c + r\alpha(\epsilon_p) - w}{c + r\alpha(\epsilon_p) - c + r\alpha(\epsilon_p)} = \frac{1}{2} - \frac{w - c}{2r\alpha(\epsilon_p)}.$$

This completes the proof. $\square$

Appendix B PROOF OF THEOREM 1

To verify that (Q_1, Q_2) satisfy condition C4, first, we show that the marginal distribution of each of the quantizers is equal to that of the corresponding LDP-FL quantizer:

$$\begin{aligned} P(Q_1(w, \mathbf{Z}) = c - r\alpha(\epsilon_p)) &= P(\mathbf{T}_1 \prec \mathbf{Z}) \\ &+ P(\mathbf{T}_1 = \mathbf{Z})P(U = -1) = 1 - \frac{1}{2^d} \lfloor 2^d q_1 \rfloor - \frac{1}{2^d} \\ &+ \frac{1}{2^d} (1 - 2^d q_1 + \lfloor 2^d q_1 \rfloor) = 1 - q_1 = \frac{1}{2} - \frac{w - c}{2r\alpha(\epsilon_p)}, \\ P(Q_2(w', \mathbf{Z}) = c - r\alpha(\epsilon_p)) &= \frac{1}{2} - \frac{w' - c}{2r\alpha(\epsilon_p)}. \end{aligned}$$

So, following the arguments in the proof of Proposition 1, conditions C4 and C5 are satisfied by (Q_1, Q_2) . To prove the bound in Equation (3), let us take an arbitrary pair of stochastic quantizers $(Q_1^*, Q_2^*) \in \mathcal{Q}$, and define

$$P_{i,j}^* = P(Q_1^*(w, \mathbf{Z}) = \gamma_i, Q_2^*(w', \mathbf{Z}) = \gamma_j),$$

where $i, j \in \{1, 2\}$. Following the arguments in the proof of Proposition 1, from condition C4, we have:

$$\begin{aligned} w &= \mathbb{E}(Q_1^*(w, \mathbf{Z})) = P(Q_1^*(w, \mathbf{Z}) = \gamma_1)\gamma_1 \\ &+ (1 - P(Q_1^*(w, \mathbf{Z}) = \gamma_1))\gamma_2 \Rightarrow P(Q_1^*(w, \mathbf{Z}) = \gamma_1) \\ &= \frac{\gamma_2 - w}{\gamma_2 - \gamma_1} \Rightarrow P_{1,1}^* + P_{1,2}^* = \frac{\gamma_2 - w}{\gamma_2 - \gamma_1}. \end{aligned} \quad (8)$$

Note that by definition:

$$P_{1,1}^* + P_{1,2}^* + P_{2,1}^* + P_{2,2}^* = 1. \quad (9)$$

Similar to Equation (8), from $w' = \mathbb{E}(Q_2^*(w', \mathbf{Z}))$ in condition C4, we have:

$$P_{1,1}^* + P_{2,1}^* = \frac{\gamma_2 - w'}{\gamma_2 - \gamma_1} \quad (10)$$

Furthermore, from condition C5, we have:

$$\begin{aligned} \frac{P(Q_1^*(w, \mathbf{Z}) = \gamma_1)}{P(Q_2^*(w', \mathbf{Z}) = \gamma_1)} &= \frac{\gamma_2 - c - r}{\gamma_2 - c + r} \leq e^{\epsilon_p} \\ \Rightarrow \gamma_2 &\geq c + r - \frac{2re^{\epsilon_p}}{e^{\epsilon_p} - 1} = c + r\alpha(\epsilon_p). \end{aligned} \quad (11)$$

Similarly $\gamma_1 \leq c - r\alpha(\epsilon_p)$. Lastly, from condition C6, the following should be minimized for all $w, w' \in [c - r, c + r]$:

$$\begin{aligned} &\mathbb{E}((Q_1^*(w, \mathbf{Z}) + Q_2^*(w', \mathbf{Z}) - w - w')^2) \\ &= \mathbb{E}((Q_1^*(w, \mathbf{Z}) - c + Q_2^*(w', \mathbf{Z}) - c - (w - c) - (w' - c))^2) \\ &= \mathbb{E}((Q_1^*(w, \mathbf{Z}) - c + Q_2^*(w', \mathbf{Z}) - c)^2 - (w + w' - 2c)^2), \end{aligned}$$

where in the last equality, we have used the unbiasedness property. Consequently, we need to minimize:

$$\begin{aligned} &\mathbb{E}((Q_1^*(w, \mathbf{Z}) - c)^2) + \mathbb{E}((Q_2^*(w', \mathbf{Z}) - c)^2) \\ &+ 2\mathbb{E}((Q_1^*(w, \mathbf{Z}) - c)(Q_2^*(w', \mathbf{Z}) - c)). \end{aligned}$$

Note that:

$$\begin{aligned} \mathbb{E}((Q_1^*(w, \mathbf{Z}) - c)^2) &= (P_{1,1}^* + P_{2,1}^*)(\gamma_1 - c)^2 \\ &+ (P_{1,2}^* + P_{2,2}^*)(\gamma_2 - c)^2. \end{aligned} \quad (12)$$

Similarly,

$$\begin{aligned} \mathbb{E}((Q_2^*(w', \mathbf{Z}) - c)^2) &= (P_{1,1}^* + P_{1,2}^*)(\gamma_1 - c)^2 + (P_{2,1}^* \\ &+ P_{2,2}^*)(\gamma_2 - c)^2. \end{aligned} \quad (13)$$

Furthermore,

$$\begin{aligned} \mathbb{E}((Q_1^*(w, \mathbf{Z}) - c)(Q_2^*(w', \mathbf{Z}) - c)) &= P_{1,1}^*(\gamma_1 - c)^2 \\ &+ (P_{1,2}^* + P_{2,1}^*)(\gamma_1 - c)(\gamma_2 - c) + P_{2,2}^*(\gamma_2 - c)^2, \end{aligned} \quad (14)$$

Consequently, we have:

$$\begin{aligned} &\mathbb{E}((Q_1^*(w, \mathbf{Z}) - c)^2) + \mathbb{E}((Q_2^*(w', \mathbf{Z}) - c)^2) \\ &+ 2\mathbb{E}((Q_1^*(w, \mathbf{Z}) - c)(Q_2^*(w', \mathbf{Z}) - c)) = P_{1,1}^*(4(\gamma_1 - c)^2) \\ &+ P_{2,2}^*(4(\gamma_2 - c)^2) + (P_{1,2}^* + P_{2,1}^*)(\gamma_1 - c + \gamma_2 - c)^2. \end{aligned}$$

Let us define $\tilde{w} = w - c$, $\tilde{w}' = w' - c$, $\tilde{\gamma}_1 = \gamma_1 - c$, and $\tilde{\gamma}_2 = \gamma_2 - c$. Then the optimization problem can be written as the minimization of:

$$P_{1,1}^*(4\tilde{\gamma}_1^2) + P_{2,2}^*(4\tilde{\gamma}_2^2) + (P_{1,2}^* + P_{2,1}^*)(\tilde{\gamma}_1\tilde{\gamma}_2)^2. \quad (15)$$

Furthermore, Equations (8) and (10) can be rewritten as:

$$P_{1,1}^* + P_{1,2}^* = \frac{\tilde{\gamma}_2 - \tilde{w}}{\tilde{\gamma}_2 - \tilde{\gamma}_1}, P_{1,1}^* + P_{2,1}^* = \frac{\tilde{\gamma}_2 - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1}.$$

We prove the theorem for the case where $|\tilde{\gamma}_1| \leq |\tilde{\gamma}_2|$. The proof for the case where $|\tilde{\gamma}_1| > |\tilde{\gamma}_2|$ follows by symmetry. We consider two subcases:

Case 1: $|\tilde{\gamma}_1| \leq |\tilde{\gamma}_2|$ and $\tilde{w} + \tilde{w}' \leq \tilde{\gamma}_1 + \tilde{\gamma}_2$:

Note that in this case, we have:

$$\frac{\tilde{\gamma}_2 - \tilde{w}}{\tilde{\gamma}_2 - \tilde{\gamma}_1} + \frac{\tilde{\gamma}_2 - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1} \geq 1.$$

Furthermore, by the assumptions of Case 1, we have $4\tilde{\gamma}_1^2 \leq 4\tilde{\gamma}_2^2$. So, Equation (15) is minimized by taking:

$$P_{1,1}^* = \frac{\tilde{\gamma}_2 - \tilde{w}}{\tilde{\gamma}_2 - \tilde{\gamma}_1} + \frac{\tilde{\gamma}_2 - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1} - 1, \quad P_{1,2}^* = 1 - \frac{\tilde{\gamma}_2 - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1}, \quad (16)$$

$$P_{2,1}^* = 1 - \frac{\tilde{\gamma}_2 - \tilde{w}}{\tilde{\gamma}_2 - \tilde{\gamma}_1}, \quad P_{2,2}^* = 0. \quad (17)$$

It should be noted that there is no guarantee that there exists a pair of stochastic quantizers (Q_1^*, Q_2^*) achieving the above distribution. Thus, minimizing the mean square error over $\tilde{\gamma}_1, \tilde{\gamma}_2$ by considering the above distribution only yields a

lower bound on the achievable mean square error. Thus, we have:

$$\begin{aligned} m(Q_1^*, Q_2^*, w, w') &\geq \min_{\tilde{\gamma}_1, \tilde{\gamma}_2} \frac{\tilde{\gamma}_1 + \tilde{\gamma}_2 - \tilde{w} - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1} (4\tilde{\gamma}_1^2) \\ &\quad + \frac{-2\tilde{\gamma}_1 + \tilde{w} + \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1} (\tilde{\gamma}_1 + \tilde{\gamma}_2)^2 \\ &= \min_{\tilde{\gamma}_1, \tilde{\gamma}_2} \frac{2\tilde{\gamma}_1^3 - 2\tilde{\gamma}_1\tilde{\gamma}_2^2 + (\tilde{w} + \tilde{w}')(-3\tilde{\gamma}_1^2 + 2\tilde{\gamma}_1\tilde{\gamma}_2 + \tilde{\gamma}_2^2)}{\tilde{\gamma}_2 - \tilde{\gamma}_1} \\ &= \min_{\tilde{\gamma}_1, \tilde{\gamma}_2} -2\tilde{\gamma}_1(\tilde{\gamma}_1 + \tilde{\gamma}_2) + (\tilde{w} + \tilde{w}')(3\tilde{\gamma}_1 + \tilde{\gamma}_2). \end{aligned} \quad (18)$$

Recall that $\tilde{\gamma}_1 \leq -r\alpha(\epsilon_p) \leq -r$, hence the derivative of Equation (18) with respect to $\tilde{\gamma}_2$ is always non-negative since $2\tilde{\gamma}_1 \leq \tilde{w} + \tilde{w}'$ for all $w, w' \in [c-r, c+r]$. So, the $\tilde{\gamma}_2$ is optimized by taking $|\tilde{\gamma}_2| = |\tilde{\gamma}_1|$ (minimum value considering the conditions in Case 1). Thus, Equation (18) simplifies to:

$$\min_{\tilde{\gamma}_1} 2(\tilde{w} + \tilde{w}')\tilde{\gamma}_1.$$

Since $\tilde{\gamma}_1 \leq 0$, and by assumption $\tilde{w} + \tilde{w}' \leq \tilde{\gamma}_1 + \tilde{\gamma}_2 = 0$, the minimum value is achieved by taking the maximum value for $\tilde{\gamma}_1$, i.e., $\tilde{\gamma}_1 = -r\alpha(\epsilon_p)$ and $\tilde{\gamma}_2 = r\alpha(\epsilon_p)$. Consequently, $\gamma_1 = c - r\alpha(\epsilon_p)$ and $\gamma_2 = c + r\alpha(\epsilon_p)$ are the optimal output values. Furthermore, from Equations (16) and (17), we have:

$$P_{1,1}^* = \frac{2c - w - w'}{2r\alpha(\epsilon_p)}, \quad P_{1,2}^* = \frac{1}{2} + \frac{w' - c}{2r\alpha(\epsilon_p)} \quad (19)$$

$$P_{2,1}^* = \frac{1}{2} + \frac{w - c}{2r\alpha(\epsilon_p)}, \quad P_{2,2}^* = 0. \quad (20)$$

To find upper-bound on the MSE of CORBINQ, we evaluate the total variation distance between $(P_{i,j})_{i,j \in \{1,2\}}$ and $(P_{i,j}^*)_{i,j \in \{1,2\}}$, where

$$P_{i,j} = P(Q_1(w, \mathbf{Z}) = \gamma_i, Q_2(w', \mathbf{Z}) = \gamma_j).$$

Note that by construction, from Algorithm 2, if $\mathbf{T}_1 \neq \mathbf{T}_2$:

$$\begin{aligned} P_{1,1} &= P(\mathbf{T}_1 \prec \mathbf{Z}, \mathbf{Z} \prec \mathbf{T}_2) + P(\mathbf{Z} = \mathbf{T}_1)P(U = -1) \\ &\quad + P(\mathbf{Z} = \mathbf{T}_2)P(U' = 1) = \frac{1}{2^d}(-\lfloor 2^d q_1 \rfloor + \lfloor 2^d(1 - q_2) \rfloor \\ &\quad - 1 + 1 - 2^d q_1 + \lfloor 2^d q_1 \rfloor + 2^d(1 - q_2) - \lfloor 2^d(1 - q_2) \rfloor) \\ &= 1 - q_1 - q_2 = P_{1,1}^*. \end{aligned}$$

Otherwise, if $\mathbf{T}_1 = \mathbf{T}_2$, then we have $P_{1,1}^* \in [0, \frac{1}{2^d}]$, and:

$$\begin{aligned} P_{1,1} &= P(\mathbf{T}_1 \prec \mathbf{Z}, \mathbf{Z} \prec \mathbf{T}_2) \\ &\quad + P(\mathbf{Z} = \mathbf{T}_1 = \mathbf{T}_2)P(U = -1)P(U' = 1) \\ &= \frac{1}{2^d}(1 - 2^d q_1 + \lfloor 2^d q_1 \rfloor)(2^d(1 - q_2) - \lfloor 2^d(1 - q_2) \rfloor) \\ &\Rightarrow 0 \leq P_{1,1} \in [0, \frac{1}{2^d}]. \end{aligned}$$

In particular, note that:

$$|P_{1,1} - P_{1,1}^*| \leq \frac{1}{2^d}.$$

Similarly, it can be observed that:

$$|P_{i,j} - P_{i,j}^*| \leq \frac{1}{2^d}, \quad i, j \in \{1, 2\}.$$

So,

$$\begin{aligned} m(Q_1, Q_2, w, w') - m(Q_1^*, Q_2^*, w, w') \\ &= \sum_{i,j \in \{1,2\}} (P_{i,j} - P_{i,j}^*)(\gamma_i + \gamma_j - w - w')^2 \\ &\leq \frac{4}{2^d} (4r\alpha(\epsilon_p))^2 = \frac{r^2 \alpha^2(\epsilon_p)}{2^{d-6}}. \end{aligned}$$

This completes the proof for Case 1.

Case 2: $|\tilde{\gamma}_1| \leq |\tilde{\gamma}_2|$ and $\tilde{w} + \tilde{w}' > \tilde{\gamma}_1 + \tilde{\gamma}_2$:

Note that in this case, we have:

$$\frac{\tilde{\gamma}_2 - \tilde{w}}{\tilde{\gamma}_2 - \tilde{\gamma}_1} + \frac{\tilde{\gamma}_2 - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1} \leq 1.$$

Furthermore, $4\tilde{\gamma}_1^2 \leq 4\tilde{\gamma}_2^2$. So, Equation (15) is minimized by taking:

$$\begin{aligned} P_{1,1}^* &= 0, \quad P_{1,2}^* = \frac{\tilde{\gamma}_2 - \tilde{w}}{\tilde{\gamma}_2 - \tilde{\gamma}_1}, \quad P_{2,1}^* = \frac{\tilde{\gamma}_2 - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1}, \\ P_{2,2}^* &= 1 - \frac{\tilde{\gamma}_2 - \tilde{w}}{\tilde{\gamma}_2 - \tilde{\gamma}_1} - \frac{\tilde{\gamma}_2 - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1}. \end{aligned}$$

Consequently, the optimization (15) can be rewritten as:

$$\begin{aligned} &\frac{-\tilde{\gamma}_1 - \tilde{\gamma}_2 + \tilde{w} + \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1} (4\tilde{\gamma}_2^2) + \frac{2\tilde{\gamma}_2 - \tilde{w} - \tilde{w}'}{\tilde{\gamma}_2 - \tilde{\gamma}_1} (\tilde{\gamma}_1 + \tilde{\gamma}_2)^2 \\ &= -2\tilde{\gamma}_2(\tilde{\gamma}_1 + \tilde{\gamma}_2) + (\tilde{w} + \tilde{w}')(3\tilde{\gamma}_2 + \tilde{\gamma}_1) \end{aligned} \quad (21)$$

The derivative with respect to $\tilde{\gamma}_1$ is always non-positive. So, $\tilde{\gamma}_1$ is optimized by taking $|\tilde{\gamma}_1| = |\tilde{\gamma}_2|$. Thus the optimization reduces to:

$$(\tilde{w} + \tilde{w}')(2\tilde{\gamma}_2),$$

which is minimized by taking $\tilde{\gamma}_1 = -r\alpha(\epsilon_p)$ and $\tilde{\gamma}_2 = r\alpha(\epsilon_p)$. Consequently, $\gamma_1 = c - r\alpha(\epsilon_p)$ and $\gamma_2 = c + r\alpha(\epsilon_p)$. The rest of the proof for Case 2 follows by similar arguments as in Case 1 and is omitted for brevity. $\square$

Appendix C

PROOF OF THEOREM 2

We have:

$$\begin{aligned} \mathbb{E}((\bar{W}_{g,j} - \frac{1}{n} \sum_{i \in [n]} w_{i,j})^2) &= \frac{1}{n^2} \sum_{i \in [n]} \text{Var}(\bar{W}_{i,j}) \\ &\quad + \frac{1}{n^2} \sum_{i, i' \in [n], i \neq i'} \text{Cov}(\bar{W}_{i,j}, \bar{W}_{i',j}). \end{aligned}$$

Following [12], we have:

$$\frac{1}{n^2} \sum_{i \in [n]} \text{Var}(\bar{W}_{i,j}) \leq \frac{r^2 \alpha^2(\epsilon_p)}{n}.$$

On the other hand:

$$\frac{1}{n^2} \sum_{i, i' \in [n], i \neq i'} \text{Cov}(\bar{W}_{i,j}, \bar{W}_{i',j}) = \frac{1}{2n^2} \sum_{i \in [n]} \text{Cov}(\bar{W}_{i,j}, \bar{W}_{p_i,j}),$$

where p_i is the index of the client paired with $\mathcal{C}_i$ as described in Algorithm 3. Furthermore,

$$\begin{aligned} \text{Cov}(\bar{W}_{i,j}, \bar{W}_{p_i,j}) &= P_{1,1}(-r\alpha(\epsilon_p))^2 \\ &\quad + (P_{1,2} + P_{2,1})(-r\alpha(\epsilon_p))(r\alpha(\epsilon_p)) + P_{2,2}(r\alpha(\epsilon_p))^2 \\ &\quad - (w_{i,j} - c)(w_{p_i,j} - c). \end{aligned}$$

Let us assume that $w_{i,j} + w_{p_i,j} \leq 2c$. Then, from Equations (19) and (20), as $d \rightarrow \infty$, we have:

$$\begin{aligned} \text{Cov}(\bar{W}_{i,j}, \bar{W}_{p_i,j}) &= \left(\frac{2c - w_{i,j} - w_{p_i,j}}{2r\alpha(\epsilon_p)} \right) (-r\alpha(\epsilon_p))^2 \\ &+ \left(1 - \frac{2c - w_{i,j} - w_{p_i,j}}{2r\alpha(\epsilon_p)} \right) (-r\alpha(\epsilon_p))(r\alpha(\epsilon_p)) \\ &- (w_{i,j} - c)(w_{p_i,j} - c) = r^2\alpha^2(\epsilon_p) \left(\frac{2c - w_{i,j} - w_{p_i,j}}{r\alpha(\epsilon_p)} - 1 \right) \\ &- (w_{i,j} - c)(w_{p_i,j} - c) \leq r^2\alpha^2(\epsilon_p) \left(\frac{2}{\alpha(\epsilon_p)} - 1 \right) - r^2 \\ &= r^2(2\alpha(\epsilon_p) - \alpha^2(\epsilon_p) - 1) = -r^2(\alpha(\epsilon_p) - 1)^2. \end{aligned}$$

Consequently,

$$\begin{aligned} \lim_{d \rightarrow \infty} \mathbb{E}((\bar{w}_{g,j} - \frac{1}{n} \sum_{i \in [n]} w_{i,j})^2) &\leq \frac{1}{n} r^2 \alpha^2(\epsilon_p) - \frac{1}{2n} r^2 (\alpha(\epsilon_p) - 1)^2 \Delta_2 = \max_{\mathbf{w}_1, \mathbf{w}'_1} \|\bar{\mathbf{W}}_1 - \bar{\mathbf{W}}'_1\|_2 \leq 2\sqrt{mr}\alpha(\epsilon_p) \\ &= \frac{1}{2n} r^2 (2\alpha^2(\epsilon_p) - (\alpha(\epsilon_p) - 1)^2) \\ &= \frac{1}{2n} r^2 ((\sqrt{2} + 1)\alpha(\epsilon_p) - 1)((\sqrt{2} - 1)\alpha(\epsilon_p) + 1). \end{aligned}$$

This completes the proof for the case where $\bar{w}_{i,j} + \bar{w}_{p_i,j} \leq 2c$. The proof for the case where $\bar{w}_{i,j} + \bar{w}_{p_i,j} > 2c$ follows by a similar argument and is omitted. $\square$

Appendix D

PROOF OF THEOREM 3

We provide an outline of the proof. Let the local update at client $\mathcal{C}_i$ be denoted by $\mathbf{w}_i \in [c - r, c + r]$, and $\bar{\mathbf{W}}_i$ be the obfuscated local update after applying the obfuscation step in AugCorBin-FL. Furthermore, let $\mathcal{T} \subseteq \mathbb{R}^m$. We wish to find (ϵ_c, δ) such that

$$P(\frac{1}{n} \sum_{i \in [n]} \bar{\mathbf{W}}_i \in \mathcal{T}) \leq e^{\epsilon_c} P(\frac{1}{n} (\bar{\mathbf{W}}'_1 + \sum_{i \in [n] - \{1\}} \bar{\mathbf{W}}_i \in \mathcal{T})) + \delta,$$

We first prove the result for the case when $\mathcal{C}_1$ is among the γ fraction of clients using the LDP-FL mechanism. Let $\mathcal{A} \subseteq [n]$ be the set of indices of clients $\mathcal{C}_i$ which belong to the γ fraction of clients using the LDP-FL mechanism. Then,

$$\sum_{i \in [n] - \{1\}} \bar{\mathbf{W}}_i = \sum_{i \in \mathcal{A} - \{1\}} \bar{\mathbf{W}}_i + \sum_{i \in \mathcal{A}^c} \bar{\mathbf{W}}_i$$

Let us define $\mathbf{B} = \sum_{i \in \mathcal{A} - \{1\}} \bar{\mathbf{W}}_i$ and $\mathbf{B}' = \sum_{i \in \mathcal{A}^c} \bar{\mathbf{W}}_i$. Then, $\mathbf{B}$ and $\mathbf{B}'$ are independent of each other. Note that for $i \in \mathcal{A}, j \in [m]$, the LDPQ quantizer generates an output $\bar{W}_{i,j}$ which is a binary with $P(\bar{W}_{i,j} = c + r\alpha(\epsilon_p)) = \frac{1}{2} + \frac{w - c}{2r\alpha(\epsilon_p)}$. Let us define $V_j = \frac{1}{2} + \frac{\bar{W}_{i,j} - c}{2r\alpha(\epsilon_p)}$. Then, $P(V_j = 1) = P(\bar{W}_{i,j} = c + r\alpha(\epsilon_p)) \in [\frac{1}{2} - \frac{1}{2\alpha(\epsilon_p)}, \frac{1}{2} + \frac{1}{2\alpha(\epsilon_p)}]$. Consequently, $\frac{1}{2} + \frac{B_j - c}{2r\alpha(\epsilon_p)}$ is Binomial with parameters $(\gamma n - 1, p)$, where $p \in [\frac{1}{2} - \frac{1}{2\alpha(\epsilon_p)}, \frac{1}{2} + \frac{1}{2\alpha(\epsilon_p)}]$. The differential privacy guarantees follow by [16]. To elaborate, following

the notation of Theorem 1 in [16], we define:

$$\begin{aligned} f(\mathcal{D}_1) &= \frac{1}{r\alpha(\epsilon_p)} (\bar{\mathbf{W}}_1 - \mathbf{c}) \in \{-1, 1\}, \\ \mathbf{Z} &= \frac{1}{2} + \frac{\mathbf{B} - \mathbf{c}}{2r\alpha(\epsilon_p)} \sim \text{BINOMIAL}(N, p), \\ N &= \gamma n - 1, \quad p = \frac{1}{2} - \frac{1}{2\alpha(\epsilon_p)}, \quad s = 2, \\ \mathcal{M}_b^{N,p,s} &= f(\mathcal{D}_1) + (\mathbf{Z} - Np)s = \frac{1}{r\alpha(\epsilon_p)} (\bar{\mathbf{W}}_1 + \mathbf{B}) \\ &- n\gamma(1 - \frac{1}{\alpha(\epsilon_p)}) + 1 - \frac{3c}{2r\alpha(\epsilon_p)}, \\ \Delta_1 &= \max_{\mathbf{w}_1, \mathbf{w}'_1} \|\bar{\mathbf{W}}_1 - \bar{\mathbf{W}}'_1\|_1 \leq 2mr\alpha(\epsilon_p) \\ \Delta_2 &= \max_{\mathbf{w}_1, \mathbf{w}'_1} \|\bar{\mathbf{W}}_1 - \bar{\mathbf{W}}'_1\|_2 \leq 2\sqrt{mr}\alpha(\epsilon_p) \\ \Delta_\infty &= \max_{\mathbf{w}_1, \mathbf{w}'_1} \|\bar{\mathbf{W}}_1 - \bar{\mathbf{W}}'_1\|_\infty \leq 2r\alpha(\epsilon_p). \end{aligned}$$

The proof then follows by substituting the above variables into Equation (7) in [16].

Similarly, if $\mathcal{C}_1$ belongs to the $(1 - \gamma)$ fraction of clients using the CorBin-FL mechanism. Then, we first note that:

$$\begin{aligned} P(\frac{1}{n} \sum_{i \in [n]} \bar{\mathbf{W}}_i \in \mathcal{T}) &= \sum_{\bar{\mathbf{w}}_{p_1} \in \mathcal{W}} P_{\bar{\mathbf{w}}_{p_1}}(\bar{\mathbf{w}}_{p_1}) P(\frac{1}{n} \sum_{i \in [n] - \{p_1\}} \bar{\mathbf{W}}_i \in \mathcal{T}'_{\bar{\mathbf{w}}_{p_1}} | \bar{\mathbf{W}}_{p_1} = \bar{\mathbf{w}}_{p_1}) \\ &= \sum_{\bar{\mathbf{w}}_{p_1} \in \mathcal{W}} P_{\bar{\mathbf{w}}_{p_1}}(\bar{\mathbf{w}}_{p_1}) P(\frac{1}{n} (\bar{\mathbf{W}}'_1 + \sum_{i \in [n] - \{1, p_1\}} \bar{\mathbf{W}}_i) \in \mathcal{T}'_{\bar{\mathbf{w}}_{p_1}}), \end{aligned}$$

where $\mathcal{W} = \{(c_j + u_j r_j \alpha(\epsilon_p))_{j \in [m]} | u_j \in \{-1, 1\}, j \in [m]\}$, $\mathcal{T}'_{\bar{\mathbf{w}}_{p_1}} = \mathcal{T} - \frac{1}{n} \bar{\mathbf{w}}_{p_1}$ is a Borel set, $\bar{\mathbf{W}}'_1$ is a random vector with distribution $P_{\bar{\mathbf{W}}_1 | \bar{\mathbf{w}}_{p_1}}(\cdot | \bar{\mathbf{w}}_{p_1})$, and in the last equality, we have used the fact that in AugCorBin-FL obfuscated updates are only pairwise dependent. Similarly,

$$\begin{aligned} P(\frac{1}{n} \sum_{i \in [n], i \neq 1} \bar{\mathbf{W}}_i \in \mathcal{T}) &= \sum_{\bar{\mathbf{w}}_{p_1} \in \mathcal{W}} P_{\bar{\mathbf{w}}_{p_1}}(\bar{\mathbf{w}}_{p_1}) P(\frac{1}{n} \sum_{i \in [n] - \{1, p_1\}} \bar{\mathbf{W}}_i \in \mathcal{T}'_{\bar{\mathbf{w}}_{p_1}}) \end{aligned}$$

Consequently, it suffices to find (ϵ_c, δ) such that for all $\bar{\mathbf{w}}_{p_1} \in \mathcal{W}$, we have:

$$\begin{aligned} P(\frac{1}{n} (\bar{\mathbf{W}}'_1 + \sum_{i \in [n] - \{1, p_1\}} \bar{\mathbf{W}}_i) \in \mathcal{T}_{\bar{\mathbf{w}}_{p_1}}) &\leq e^{\epsilon_c} P(\frac{1}{n} \sum_{i \in [n] - \{1, p_1\}} \bar{\mathbf{W}}_i \in \mathcal{T}_{\bar{\mathbf{w}}_{p_1}}) + \delta. \end{aligned}$$

The privacy guarantees then follow by applying Theorem 1 in [16] as in the previous case. The mean square error guarantees follow directly from Proposition 1 and Theorem 1.

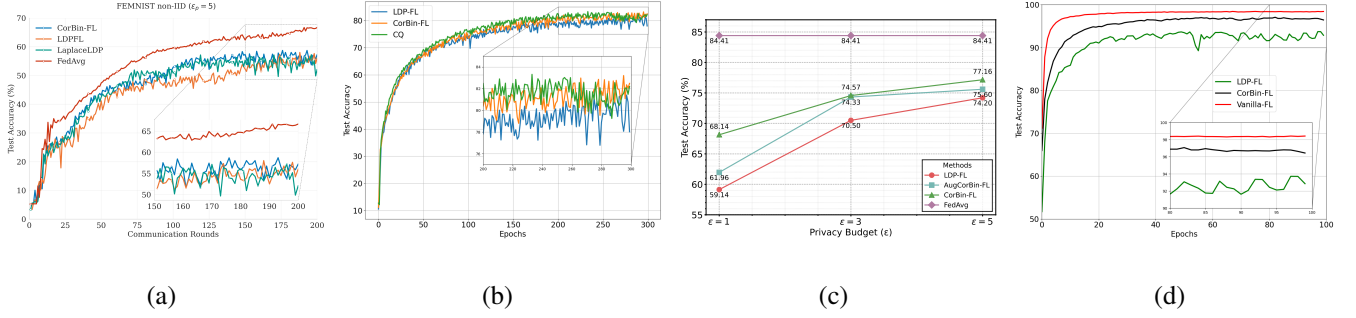


FIGURE 4. Additional experimental results evaluating (a) Test accuracy on the FEMNIST, non-IID, $\epsilon_p = 5$, (b) the accuracy performance of LDP-FL, CorBin-FL, and CQ [21], (c) performance of LDP-FL, AugCorBin-FL, and CorBin-FL under various PLDP privacy budgets, and (d) accuracy performance of LDP-FL and CorBin-FL on the MNIST dataset.

Appendix E Proof of Theorem 4

We provide an outline of the proof. Let $\mathcal{P}_i, i \in [\frac{n}{2}]$ be the collection of pairings of the clients in CorBin-FL, and let E_i be the indicator that exactly one of the two clients in the i th pair drops out. Then,

$$P(E_i = 1) = 2p(1-p), \quad \mathbb{E}(\sum_{i \in [\frac{n}{2}]} E_i) = np(1-p)$$

$$\text{Var}(\sum_{i \in [\frac{n}{2}]} E_i) = np(1-p)(1-2p(1-p)).$$

Consequently, using the Chebychev's inequality, we have:

$$P(\sum_{i \in [\frac{n}{2}]} E_i \in [\gamma n, \gamma' n]) \geq 1 - (\delta - \delta_1)$$

Let $\mathcal{E}$ represent the event that $\sum_{i \in [\frac{n}{2}]} E_i \in [\gamma n, \gamma' n]$. Then,

$$P(\sum_{i \in [n]} \bar{W}_i \leq \mathcal{T}) \leq P(\mathcal{E})P(\sum_{i \in [n]} \bar{W}_i \leq \mathcal{T} | \mathcal{E}) + P(\mathcal{E}^c)$$

$$\leq P(\sum_{i \in [n]} \bar{W}_i \leq \mathcal{T} | \mathcal{E}) + (\delta - \delta_1)$$

$$\leq e^{\epsilon_c} P(\sum_{i=2}^n \bar{W}_i \leq \mathcal{T} | \mathcal{E}) + \delta_1 + \delta - \delta_1,$$

where the last inequality follows from the proof of Theorem 3. This completes the proof of the UCDP guarantee. Furthermore, let $N = \sum_{i \in [n]} F_i$, where F_i is the indicator that the i th client does not dropout. Note that:

$$P(F_i = 1) = p, \quad \mathbb{E}(\sum_{i \in [n]} F_i) = np, \quad \text{Var}(\sum_{i \in [n]} F_i) = np(1-p).$$

Then,

$$P(\sum_{i \in [n]} F_i \in [\theta n, \theta' n]) \geq 1 - (\delta - \delta_1),$$

where $\theta = p(1-p) - \sqrt{\frac{p(1-p)}{(\delta-\delta_1)n}}$ and $\theta' = p(1-p) + \sqrt{\frac{p(1-p)}{(\delta-\delta_1)n}}$. Let $\mathcal{F}$ be the event that $\sum_{i \in [n]} F_i \in [\theta n, \theta' n]$,

then:

$$\lim_{d \rightarrow \infty} \mathbb{E}((\bar{W}_{g,j} - \frac{1}{N} \sum_{i: F_i=1} w_{i,j})^2)$$

$$\leq \lim_{d \rightarrow \infty} \mathbb{E}((\bar{W}_{g,j} - \frac{1}{N} \sum_{i: F_i=1} w_{i,j})^2 | \mathcal{E} \cap \mathcal{F})$$

$$+ (2r\alpha(\epsilon_p))^2 (P(\mathcal{E}^c) + P(\mathcal{F}^c))$$

$$\leq \frac{(1-\gamma)}{2n\theta} r^2 ((\sqrt{2}-1)\alpha(\epsilon_p) + 1) ((\sqrt{2}+1)\alpha(\epsilon_p) - 1)$$

$$+ \frac{\gamma' r^2 \alpha^2(\epsilon_p)}{n\theta} + 8r^2 \alpha^2(\epsilon_p) \delta_1.$$

This completes the proof. $\square$

Appendix F ADDITIONAL EXPERIMENTAL RESULTS

a: Correlated Quantization without Privacy (Figure 4b)

CorBin-FL achieves low communication cost via one-bit quantization. To benchmark its accuracy under weak privacy, we compare it with the correlated quantization (CQ) method from [21], which does not ensure privacy. We use the CIFAR-10 dataset with 50 clients (batch size 64), setting $\epsilon_p = 10$ and $d = 5$ for CorBin-FL, and $\lambda = 1$ across all methods. We also include LDP-FL with $\epsilon_p = 50$ as a relaxed-privacy baseline. As shown in Figure 4b, CorBin-FL achieves nearly the same accuracy as CQ ($\epsilon_p \rightarrow \infty$) and LDP-FL with $\epsilon_p = 50$.

b: Experiment on FEMNIST Non-IID (Figure 4a)

We evaluate CorBin-FL on the non-IID FEMNIST dataset from the LEAF benchmark [27], where data heterogeneity arises from writer-specific labels. Using 350 clients over 200 rounds, a CNN model (Appendix H) with local SGD is trained, employing adaptive learning rates and stabilization for non-IID data. Test accuracy and privacy-utility trade-offs for $\epsilon_p = 5$ are shown in Figure 4a.

c: Accuracy for Varying Privacy Budgets (Figure 4c)

We compare the accuracy of CorBin-FL, AugCorBin-FL, and LDP-FL on CIFAR-10 across privacy budgets $\epsilon_p \in 1, 3, 5$, using a grid search over $\lambda \in 0.1, 0.2, \dots, 1.0$ for each ϵ_p . Figure 4c shows the best accuracy for each method.

d: Experiments on the MNIST Dataset (Figure 4d)

We evaluate CorBin-FL on MNIST using a two-layer VGG model under a fixed privacy budget of $\epsilon_p = 0.5$, comparing it with LDP-FL and Vanilla-FL (no privacy). A grid search over $\lambda \in 0.1, \dots, 1.0$ is performed for all methods (Figure 4d). CorBin-FL outperforms LDP-FL, with the gap increasing at lower ϵ_p . At $\epsilon_p = 0.5$, its accuracy is within 1.5% of Vanilla-FL.

e: MSE comparison of LDPQ and CorBinQ (Figure 5)

We focus on the quantizers used in the CorBin-FL and LDP-FL mechanisms, namely the CorBinQ and LDPQ quantizers. We plot the resulting MSE when feeding a pair of quantizers with two input weights $w_1, w_2 \in [c - r, c + r]$, where we have taken $c = 0$ and $r = 0.5$. Figure 5 shows the MSE comparison between a pair of clients utilizing CorBinQ in Algorithm 2 and a pair of clients using the LDPQ in Algorithm 1. The figure demonstrates the MSE gains of CorBinQ, particularly at smaller values of the privacy budget ϵ_p . There is a larger gap in variance between CorBinQ and LDPQ methods as ϵ_p decreases. This suggests that correlated randomness is especially beneficial in scenarios requiring stronger privacy guarantees (i.e., smaller ϵ_p values).

Appendix G

LIMITATIONS OF THE PROPOSED PRIVACY MECHANISMS

While CorBin-FL and AugCorBin-FL show strong theoretical and empirical performance—demonstrating robustness to dropout, scalability, and effectiveness in both IID and non-IID FL settings—a key limitation is the need to share several bits of common randomness among clients per communication round. This requirement is still less stringent than that in recent works such as [18], [19], which rely on correlated Gaussian noise and demand significantly more shared bits per model parameter.

We use secure key exchange protocols to enable randomness sharing, though alternate approaches can extract correlated randomness from shared observations (e.g., common environments or physical wireless channels) as explored in [28], [29]. Analyzing the impact of replacing common with correlated randomness on privacy and utility remains an open and promising direction.

Appendix H

MODEL ARCHITECTURES

Tables 5 and 6 detail the neural network architectures used in our experiments. For all convolutional models, we use ReLU activation functions after each layer unless otherwise specified. The variable k denotes kernel size and p denotes padding.

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TABLE 5. Vision Model Architectures

DATASET	ARCHITECTURE DETAILS
MNIST (VGG-STYLE)	INPUT: 1×28×28 CONV2D(1→32, K=3, P=1) + MAXPOOL2D(2×2) CONV2D(32→64, K=3, P=1) + MAXPOOL2D(2×2) DENSE(3136→128) + DROPOUT(0.5) + DENSE(128→10)
CIFAR10 (RESNET18)	INPUT: 3×32×32 CONV2D(3→64, K=3, P=1) + BATCHNORM + RELU 2× BASICBLOCK(64→64) 2× BASICBLOCK(64→128, STRIDE=2) 2× BASICBLOCK(128→256, STRIDE=2) 2× BASICBLOCK(256→512, STRIDE=2) AVGPOOL2D(4×4) + DENSE(512→10)
FEMNIST	INPUT: 1×28×28 CONV2D(1→32, K=5, P=2) + MAXPOOL2D(2×2) CONV2D(32→64, K=5, P=2) + MAXPOOL2D(2×2) DENSE(3136→2048) + DENSE(2048→62)

TABLE 6. Text Model Architectures

DATASET	ARCHITECTURE DETAILS
SHAKESPEARE	EMBEDDING: EMBED(80→8) LSTM: HIDDEN SIZE=256, NUM LAYERS=2 OUTPUT: DENSE(256→80)
SENTIMENT140	EMBEDDING: EMBED(VOCAB_SIZE→300) LSTM: HIDDEN SIZE=100, NUM LAYERS=2 OUTPUT: DENSE(100→128) + DROPOUT(0.3) + DENSE(128→2)

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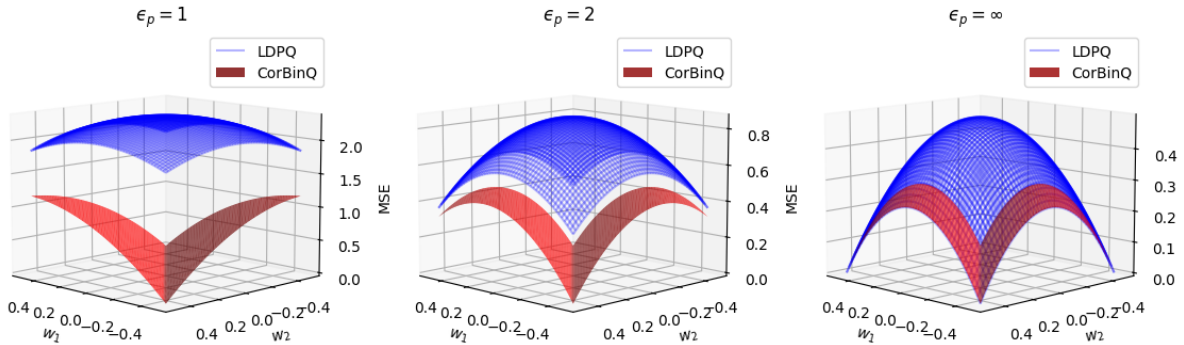


FIGURE 5. MSE of CorBinQ and LDPQ pairs of quantizers for inputs $w_1, w_2 \in [-0.5, 0.5]$.

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